Lecture 10: Operators in C

Lesson objectives

By the end of this lesson, you will be able to:

- **Define** the concept of operators
- **Describe** various ways of classifying operators.
- **Differentiate** between associativity and order of precedence.
- Given some expression/formula **explain** the associativity and order of precedence
- **Chose** operators for a formula and/or expression.

Next lesson **10/03/2026**

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Operator=symbol/object that define the type of calculation to be performed on operands

Operand=value to calculate e.g a+b , a and b are operands.

- Order of **precedence** = priority for operators.

Associativity= Direction of execution

Expression= statement that can evaluate to a value.

Statement= a complete instruction that tells the CPU what to do.

- How to classify operators?
- **A) By number of operands**
  - **Unary operators-** operator works on one operand e.g. ++,--
  - **Binary operators-** operator works on two operands e.g. +,-,/,*
  - **Ternary operators-** operator works on three operands e.g. ?:

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B) By type of calculations

Operators	Examples	Usage
Arithmetic	+,*,/,-,%,--,++	Return numeric results
Relational	==,>,<,>=,<=,!=	Return logical result
Logical	&&,||,!	Combines relational expressions
Assignment	=,+=,-=,/=,*=	Assigns values to variables
Miscellaneous	sizeof,&,?:, *, [] , .	Have special meaning
Bitwise	>>,<<,&,|,^	Used to manipulate bits

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Binary manipulations

X	=	6			(00000110)									
Y	=	7			(00000111)									
X	>>	2			0	0	0	0	0	1	1	0		
X	<<	2					0	0	0	0	0	1		
X	&	Y	0	0	0	0	0	1	1	0	0	0		
X	|	Y												
X	^	Y												
						0	0	0	0	0	1	1	0	
						0	0	0	0	0	1	1	1	
						0	0	0	0	0	0	0	1	

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```c
#include<stdio.h>
int main()
{
    int x,y;
    printf("Enter value of x and y\n");
    scanf("%d%d",&x,&y);
    (x>y)?printf("%d is greater\n",x):printf("%d is greater\n",y);
    printf("%p is the address of x\n",&x);
    printf("%d bytes is the memory size of x\n",sizeof(x));
    return 0;
}
```

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Associativity:

- Associativity is the rule that determines the direction in which operators of the same precedence are evaluated.
- It is used when an expression contains multiple operators with the same precedence.
- It ensures that expressions are evaluated consistently and helps programmers predict the correct result of calculations.

Example: Consider this table

Expression	**How it is evaluated**
A-B-C	(A-B)-C //Left-Right
X/Y/Z	(X/Y)/Z //Left-Right
A=B=C=8	A=(B=(C=8)) //Right-Left

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Order of precedence:

- Order of precedence is the rule that determines which operator is evaluated first in an expression.
- Precedence decides which operator comes first.
- **Example:** Consider this table

Expression	**How it is evaluated**
A+B*C	A+(B*C) //Multiply first then add
(X+Y)*Z	(X+Y)*Z //Evaluate brackets first then multiply
A<B && C>D	Evaluate (A<B) first, then (C>D) finally evaluate (A<B) &&(C>D)

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Summary:

- This lecture introduced operators as symbols used to perform operations on data in C programs.
- It explored different classifications of operators, including arithmetic, relational, logical, assignment, and bitwise operators.
- The lecture also distinguished between associativity and precedence, explaining how they determine the order in which expressions are evaluated.
- Finally, learners practiced analyzing expressions and selecting appropriate operators to construct correct formulas and program statements.

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Activity:

Write a C program to initialize four variables a,b,x and y then perform the following.

a) Check if either (a+b) or (x+y) are smaller than 15

b) Check if either a or b is smaller than 12 and either x or y is larger than 8

c) Check if a is larger than equal to x and b is larger than equal to y or x is smaller than 100 and y is greater than 8

d) Check if (a|b) is not equal to (x^y)

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Exercise

a) Explain the output of the following program

```
#include<stdio.h>
int main(){
int a=5,b=-2;
int x=(a>=0)&&(b<-a);
printf("x=%d",x);
return 0;
}
```

b) Given the following operators, arrange them in ascending order of priority

%,--,/,*,++,+,-

References

1. An introduction to programming by Wu Thomas, Norman and Theodore: McGrill (1997).
2. C programming language by Brian W. Kernighan, Dennis Ritchie, 2nd ed: prentice hall(2003).
3. C how to program by H.M. Deitel and P.J.Deitel, 3rd ed: Prentice hall(2001).